

# HE2003 Econometrics I: Linear Regression with Multiple Regressors 5

**Wenjie Wang**

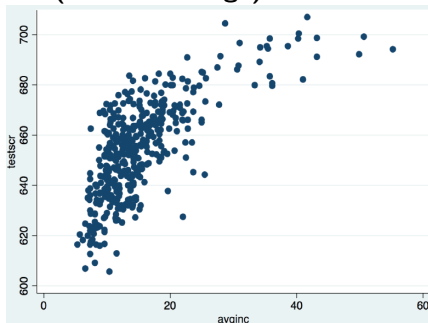
**Economics, School of Social Sciences  
Nanyang Technological University**

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# Incorporating Nonlinearities in Linear Regression

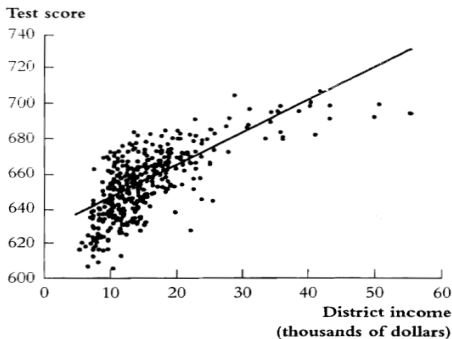
- Everything so far has been linear in  $X$  and  $Y$ . But the linear model is not always a good one.
- For example, we expect that **students' test score increases when parents' income increases**. The scatterplot confirms our expectation. We also observe that the **relationship is nonlinear**: The slope becomes flatter as income increases.

**Scatterplot of (district-average) test scores and incomes**



# Incorporating Nonlinearities in Linear Regression

- As shown in the following graph, the **simple linear regression does not fit well the nonlinear relationship** in the test score - income data.



- A useful way to **specify a nonlinear relationship** is to use the **natural logarithm** of  $X$  and/or  $Y$ : Logarithms convert changes in variables into **percentage changes**, and many relationships in economics are naturally expressed in terms of percentages.

# Incorporating Nonlinearities in Linear Regression

## KEY CONCEPT: Linear-log Model

- For simplicity, we consider the model with one explanatory variable. The result **can be extended directly to multiple regression** model.
- In the linear-log case, the regression model is

$$Y_i = \beta_0 + \beta_1 \log(X_i) + u_i, i = 1, \dots, n.$$

where  $\log(X_i)$  is the **natural logarithm** of  $X_i$  [also can be written as  $\ln(X_i)$ ]. The only difference between this model and the simple linear regression we have already learnt is that the **right-hand-side variable is now  $\log(X_i)$  rather than  $X_i$  itself.**

- **The meaning of  $\beta_1$ :**

$$\beta_1 \approx \frac{\Delta Y}{\Delta X/X} = \frac{\text{change in } Y}{\text{percentage change in } X}.$$

For example,  $X = 10$  and  $\Delta X = 1$ , then  $\Delta X/X = 1/10 = 10\%$ . Suppose  $\beta_1 = 2$  in this case, then the corresponding change in  $Y$  is:  $\Delta Y \approx \beta_1 \times \Delta X/X = 2 \times 10\% = 0.2$ .

# Incorporating Nonlinearities in Linear Regression

## Derivation. (need not recite)

Consider the population regression function:

$$Y = \beta_0 + \beta_1 \log(X) \quad (1)$$

$$Y + \Delta Y = \beta_0 + \beta_1 \log(X + \Delta X) \quad (2)$$

Subtract eq.(2) - eq.(1):

$$\Delta Y = \beta_1 [\log(X + \Delta X) - \log(X)] \quad (3)$$

Now, we use the following **approximation** (when  $\frac{\Delta X}{X}$  is small):

$$\log(X + \Delta X) - \log(X) \approx \frac{\Delta X}{X}, \quad (4)$$

which comes from calculus:  $d\log(X) = \frac{1}{X} dx$ . For example,  
 $\log(40 + 1) - \log(40) = 0.0247$  vs.  $1/40 = 0.025$ ;  
 $\log(80 + 1) - \log(80) = 0.0124$  vs.  $1/80 = 0.0125$ .

Combining eq.(3) and eq.(4), we obtain:

$$\Delta Y \approx \beta_1 \frac{\Delta X}{X}, \text{ and } \beta_1 \approx \frac{\Delta Y}{\Delta X / X}.$$

# Incorporating Nonlinearities in Linear Regression

## Example: Test Score and Income

Now we want to apply the linear-log model to the California data set.

- First, define the **new regressor**,  $\log(\text{Avg\_Income})$ .
- The model is now linear in  $\log(\text{Avg\_Income})$ , so the **linear-log model can be estimated by OLS**:

$$\widehat{\text{TestScore}} = 557.8 + 36.42 \log(\text{Avg\_Income}), R^2 = 0.56.$$

(3.8)      (1.40)

A **1% increase in district-average income** is approximately associated with an increase in test score of  **$36.42 \times 1\% = 0.3642$  point** (pay attention to the unit):

$$\text{Predicted change in } Y = \widehat{\Delta Y} \approx \hat{\beta}_1 \times \frac{\Delta X}{X} = 36.42 \times 1\% = 0.3642.$$

- Standard errors, t-test, p-value, confidence interval, and  $R^2$  - **all the usual tools of linear regression apply here.**

# Incorporating Nonlinearities in Linear Regression

**STATA command for linear-log model:**

1. **gen log\_avg\_income = log(avginc)**

[The command “gen” generates a new variable as a function of original variable. Here, we use the logarithm function log(.).]

2. **reg testscr log\_avg\_income, robust**

```
. gen log_avg_income = log(avginc)

. reg testscr log_avg_income, robust
```

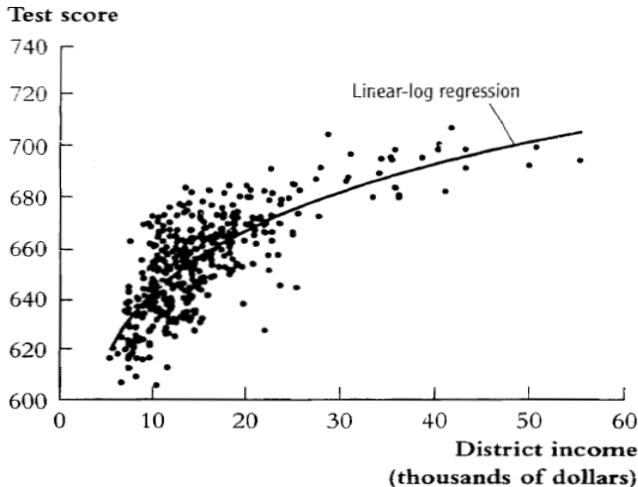
Linear regression

```
Number of obs =      420
F( 1, 418) =    679.70
Prob > F       =    0.0000
R-squared      =    0.5625
Root MSE      =    12.618
```

| testscr        | Robust   |           | t      | P> t  | [95% Conf. Interval] |          |
|----------------|----------|-----------|--------|-------|----------------------|----------|
|                | Coef.    | Std. Err. |        |       |                      |          |
| log_avg_income | 36.41968 | 1.396943  | 26.07  | 0.000 | 33.67378             | 39.16559 |
| _cons          | 557.8323 | 3.83994   | 145.27 | 0.000 | 550.2843             | 565.3803 |

## Incorporating Nonlinearities in Linear Regression

The linear-log model fits quite well the test score - income data.





## Incorporating Nonlinearities in Linear Regression

**Example: Testing the “Efficient Market Hypothesis” in Financial Economics (continued)** Now, we re-estimate the model by using the **log** form of “netinc” and “salary”:

$$\begin{aligned} & \widehat{\text{return}} \\ = & -36.30 + 0.327dkr + 0.069eps - 4.74\log(\text{netinc}) + 7.24\log(\text{salary}), \\ & (39.37) \quad (0.203) \quad (0.080) \quad (3.39) \quad (6.31) \\ & n = 142, R^2 = 0.0330. \end{aligned}$$

**Question:** What are the interpretations of the OLS coefficients for  $\log(\text{netinc})$  and  $\log(\text{salary})$ ? **Overall, is the evidence for predictability of stock returns strong or weak under this model?**

**Answer:**  $\hat{\beta}_3 = -4.74$ ,  $\beta_3 \approx \frac{\Delta Y}{\Delta X_3/X_3}$ ,  $\widehat{\Delta Y} \approx \hat{\beta}_3 \cdot \frac{\Delta X_3}{X_3} = -4.74 \cdot 1\%$   
 $= -0.0474$  (dollar, assuming the unit of return is dollar). Therefore, given 1% change of *netincome*, *return* is (approximately) predicted to change by  $-0.0474$  (dollar). Similarly,  $\hat{\beta}_4 = 7.24$ ,  $\beta_4 \approx \frac{\Delta Y}{\Delta X_4/X_4}$ ,  
 $\widehat{\Delta Y} \approx \hat{\beta}_4 \cdot \frac{\Delta X_4}{X_4} = 7.24 \cdot 1\% = 0.0724$  (dollar). Given 1% change of *salary*, *return* is (approximately) predicted to change by 0.0724 (dollar).

# Incorporating Nonlinearities in Linear Regression

## KEY CONCEPT: Log-log Model

- We may also specify both  $X$  and  $Y$  in terms of percentages. This is called the log-log model. In the log-log case, the regression model is:

$$\log(Y_i) = \beta_0 + \beta_1 \log(X_i) + u_i.$$

- The meaning of  $\beta_1$ :

$$\beta_1 \approx \frac{\Delta Y/Y}{\Delta X/X} = \frac{\text{percentage change in } Y}{\text{percentage change in } X}$$

Therefore,  $\beta_1$  is, in fact, the elasticity of  $Y$  with respect to  $X$ . (e.g.,  $X$  is price and  $Y$  is consumer demand, where  $\beta_1$  is negative).

For example,  $\Delta X/X = 1/10 = 10\%$ . Again, suppose  $\beta_1 = 2$  in this case, then the corresponding percentage change in  $Y$  is:

$$\Delta Y/Y \approx \beta_1 \times \Delta X/X = 2 \times 10\% = 0.2 = 20\%.$$

That is, when  $X$  changes 10%, the corresponding percentage change in  $Y$  is 20% (Not 0.2 point, because here we are looking at percentage change. This is different from the linear-log model, which looks at change in the unit of  $Y$ ).

# Incorporating Nonlinearities in Linear Regression

## Example: Test Score and Income (cont'd)

- First, define the **new dependent variable**  $\log(\text{TestScore})$  and the **new regressor**  $\log(\text{Avg\_Income})$ .
- The model is now a linear regression of  $\log(\text{TestScore})$  on  $\log(\text{Avg\_Income})$ , so it **can be estimated by OLS**:

$$\log(\widehat{\text{TestScore}}) = 6.336 + 0.0554\log(\text{Avg\_Income}), \quad R^2 = 0.557.$$

(0.006)      (0.002)

According to this result, a **1% increase in income** is estimated to correspond to a **0.0554% increase in test scores**:

**Predicted percentage change in Y**

$$= \widehat{\Delta Y/Y} \approx \hat{\beta}_1 \times \frac{\Delta X}{X} = 0.0554 \times 1\% = 0.0554\%.$$

- For example, suppose the **district-average income increases from \$10,000 to \$11,000, or by 10%**. Then test score increases by approximately  $0.0554 \times 10\% = 0.554\%$ .

# Incorporating Nonlinearities in Linear Regression

STATA command for log-log model:

1. `gen log_testscr = log(testscr)`
2. `gen log_avg_income = log(avginc)`
3. `reg log_testscr log_avg_income, robust`

```
. gen log_testscr = log(testscr)

. gen log_avg_income = log(avginc)

. reg log_testscr log_avg_income, robust
```

Linear regression

```
Number of obs =    420
F( 1, 418) =    667.78
Prob > F      =    0.0000
R-squared     =    0.5578
Root MSE     =    .01938
```

| log_testscr    | Robust   |           |         |       |                      |          |
|----------------|----------|-----------|---------|-------|----------------------|----------|
|                | Coef.    | Std. Err. | t       | P> t  | [95% Conf. Interval] |          |
| log_avg_income | .055419  | .0021446  | 25.84   | 0.000 | .0512035             | .0596345 |
| _cons          | 6.336349 | .0059246  | 1069.50 | 0.000 | 6.324704             | 6.347995 |

# Incorporating Nonlinearities in Linear Regression

**Empirical Application: The Colonial Origins of Comparative Development**  
Acemoglu, Johnson and Robinson (AJR, American Economic Review, 2001) are interested in the **effect of political institutions on prosperity and economic performance**: e.g., **good institution** (strong rule-of-law, protection of property rights) should result in a country having **higher long-term economic growth**. They use a sample of **64 former European colonies** and argue that European colonizing countries tended to set up colonies as either an **“extractive state”** or as a **“migrant colony”**.

The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 2024

Daron Acemoglu

“for studies of how institutions are formed and affect prosperity”



Daron Acemoglu. Ill. Niklas Elmehed © Nobel Prize Outreach

Simon Johnson

“for studies of how institutions are formed and affect prosperity”



Simon Johnson. Ill. Niklas Elmehed © Nobel Prize Outreach

James Robinson

“for studies of how institutions are formed and affect prosperity”



James Robinson. Ill. Niklas Elmehed © Nobel Prize Outreach

<https://www.nobelprize.org/prizes/economic-sciences/2024/press-release/>

## Incorporating Nonlinearities in Linear Regression

AJR highlighted that **initial conditions** in areas colonized by Europeans differed largely by disease environments, in particular for the settlers themselves. They hypothesized that Europeans settled to a greater extent in **places where their mortality rates were relatively low**; they brought along **their values as well as beliefs and developed home-like institutions**, in particular economic and political institutions benefitting themselves. **In areas where mortality was high** due to diseases such as malaria and yellow fever, Europeans did not settle in large numbers. In these places, therefore, the colonizers instead **established or maintained institutions designed to extract resources** from native populations as much and as quickly as possible. Many characteristics of the colonial state and institutions persisted even after independence, and that such persistence **influences contemporary economic performance**.

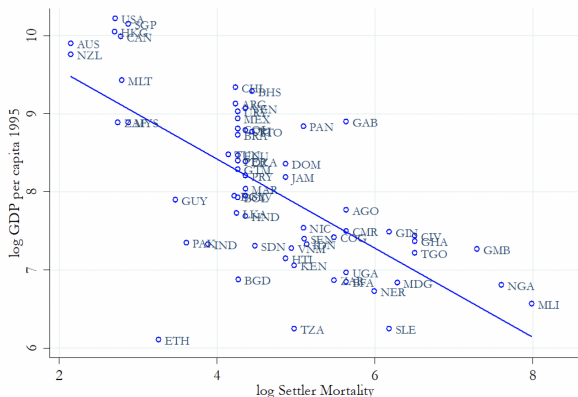
**Figure 2:** Proposed causal chain in Acemoglu, Johnson, and Robinson (2001)

1. Living conditions during colonization (settler mortality)  $\Rightarrow$  2. Size of European settlements during colonization  $\Rightarrow$  3. Colonial institutions  $\Rightarrow$  4. Institutions at independence  $\Rightarrow$  5. Contemporary institutions  $\Rightarrow$  6. Contemporary economic prosperity

# Incorporating Nonlinearities in Linear Regression

**Figure 3.** Settler mortality, institutions, and prosperity

Panel A

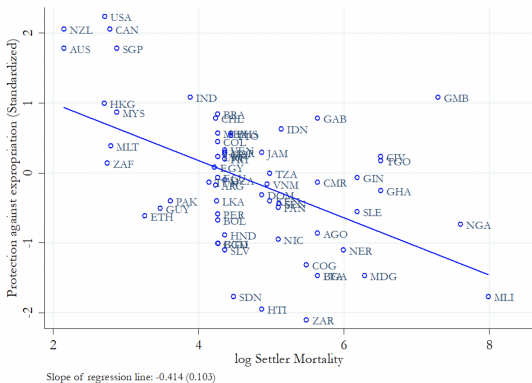


Slope of regression line: -0.570 (0.073)

**Question: Construct a log-log model for GDP per capita (1995) and Settler Mortality (19th century), and explain the meaning of the slope estimate -0.57.**

# Incorporating Nonlinearities in Linear Regression

Panel B



**Notes.** Panel A: Relationship between log GDP per capita in 1995 (purchasing power adjusted) and log of settler mortality and. Panel B: Relationship between average protection against expropriation risk 1985-95 and log of settler mortality. Panels A and B are analogous to Figures 1 and 3 in Acemoglu, Johnson, and Robinson (2001). They have been reproduced using data from Acemoglu, Johnson, and Robinson (2012).

**Question: Construct a linear-log model for Protection Against Expropriation Risk (1985-95) and Settler Mortality (19th century), and explain the meaning of the slope estimate -0.414.**



# Incorporating Nonlinearities in Linear Regression

## KEY CONCEPT: Log-linear Model

- In the log-linear case, the regression model is

$$\log(Y_i) = \beta_0 + \beta_1 X_i + u_i.$$

- The meaning of  $\beta_1$ :

$$\beta_1 \approx \frac{\Delta Y/Y}{\Delta X} = \frac{\text{percentage change in } Y}{\text{change in } X}$$

For example,  $\Delta X = 2$ . Suppose  $\beta_1 = 0.2$  in this case, then the corresponding percentage change in  $Y$  is:

$$\Delta Y/Y \approx \beta_1 \times \Delta X = 0.2 \times 2 = 0.4 = 40\%.$$

That is, when  $X$  changes by 2 units, the corresponding percentage change in  $Y$  is 40%.

# Incorporating Nonlinearities in Linear Regression

**Question:** Consider the following regression model:

$$\widehat{\log(\text{price})} = 9.23 - 0.718\log(\text{nox}) + 0.306\text{rooms},$$

(0.19) (0.066) (0.019)

$$n = 506, R^2 = 0.514,$$

where *price* refers to the price of apartment, *nox* refers to the amount of pollution, and *rooms* refers to the number of rooms in the apartment. Standard errors are reported below the estimates.

- Explain the meaning of -0.718.
- Explain the meaning of 0.306.
- Test the null hypothesis that neither *nox* nor *rooms* has a significant effect on *price* using the homoskedasticity-only *F*-test.

# Incorporating Nonlinearities in Linear Regression

## Summary of Different Models:

1. **Linear Regression Model:**  $Y_i = \beta_0 + \beta_1 X_i + u_i$ ; Interpretation of  $\beta_1$ :

$$\beta_1 = \frac{\Delta Y}{\Delta X} = \frac{\text{change in } Y}{\text{change in } X}$$

2. **Linear-log Model:**  $Y_i = \beta_0 + \beta_1 \log(X_i) + u_i$ ; Interpretation of  $\beta_1$ :

$$\beta_1 \approx \frac{\Delta Y}{\Delta X/X} = \frac{\text{change in } Y}{\text{percentage change in } X}$$

3. **Log-log Model:**  $\log(Y_i) = \beta_0 + \beta_1 \log(X_i) + u_i$ ; Interpretation of  $\beta_1$ :

$$\beta_1 \approx \frac{\Delta Y/Y}{\Delta X/X} = \frac{\text{percentage change in } Y}{\text{percentage change in } X}$$

4. **Log-linear Model:**  $\log(Y_i) = \beta_0 + \beta_1 X_i + u_i$ ; Interpretation of  $\beta_1$ :

$$\beta_1 \approx \frac{\Delta Y/Y}{\Delta X} = \frac{\text{percentage change in } Y}{\text{change in } X}$$